

		rate of change of gravitational potential energy) = $(200)(9.81)(6) = 11760 \text{ W}$ % efficiency = (useful power/ output power) x 100% = $(7360/15696) \times 100\% = 47\%$
10	B	$N \cos 40^\circ = mg$ -----eqn (1) $N \sin 40^\circ = mv^2/r$ ----- eqn (2) Eqn (2)/ eqn (1), $\tan 40^\circ = v^2/rg$ $r = 34^2 / (\tan 40^\circ)(9.81) = 140.43 \text{ m}$ $d = r / \cos 40^\circ = 183.32 = 183 \text{ m}$
11	D	A small wheel of radius a is free to rotate about a horizontal axis through its center, perpendicular to the page. A force F is applied to the rim of the wheel, as shown. The wheel is released from rest in the position shown. The angular displacement of the wheel as it rotates through an angle ϕ is given by $\phi = \frac{1}{2} \alpha t^2$, where α is the angular acceleration. The angular velocity of the wheel at the instant it has rotated through an angle ϕ is $\omega = \alpha t$. The angular displacement of the wheel as it rotates through an angle ϕ is given by $\phi = \frac{1}{2} \alpha t^2$, where α is the angular acceleration. The angular velocity of the wheel at the instant it has rotated through an angle ϕ is $\omega = \alpha t$.